

A Curvature-Threshold Scalar Topographic EFT as a Mass-Gap Analogue of Yang–Mills Theory

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Abstract

The curvature-threshold scalar topographic effective field theory (EFT) provides a minimal, analytically tractable framework in which a rigorously demonstrable mass gap arises from internal geometric structure. This manuscript consolidates the mass-gap theorem, the theoretical framework, and a structural comparison to the Yang–Mills mass-gap problem. The model exhibits a localized nonlinear background, a positive elliptic fluctuation operator, a discrete spectrum with a strictly positive lowest eigenvalue, and a geometric origin of mass. These features mirror the mathematical architecture required for a Yang–Mills mass-gap proof and position the curvature-threshold EFT as a conceptual analogue and potential pathway toward resolving aspects of the Clay Millennium Problem.

1 Introduction

The Yang–Mills mass-gap problem is one of the seven Clay Millennium Prize Problems. It asks for a rigorous demonstration that quantum Yang–Mills theory in four dimensions possesses a mass gap: a strictly positive lower bound on the excitation spectrum above the vacuum. Despite decades of progress, no complete mathematical proof exists. The difficulty lies in constructing a nonperturbative framework yielding a self-adjoint Hamiltonian with a discrete spectrum and a strictly positive lowest eigenvalue.

This manuscript develops a scalar topographic EFT based on a curvature-threshold mechanism. The model exhibits a rigorously provable mass gap and mirrors the mathematical structure required for a Yang–Mills mass-gap proof. Its geometric clarity and analytical tractability make it a compelling analogue for exploring nonperturbative spectral structure.

2 Theoretical Framework

We consider a real scalar field theory in $3 + 1$ dimensions with scalar density

$$\rho(x, t) = e^{\phi(x, t)}.$$

The curvature operator is defined by

$$K[\rho] = -\nabla^2 \ln \rho = -\nabla^2 \phi.$$

A localized background configuration $\phi_0(x)$ satisfies the curvature-activation condition

$$-\nabla^2 \phi_0 = k_{\text{loc}},$$

where $k_{\text{loc}} > 0$ is the effective local curvature scalar.

The Lagrangian density is

$$L = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}(\nabla \phi)^2 - \frac{\lambda}{2}(-\nabla^2 \phi - k_{\text{loc}})^2,$$

with $\lambda > 0$ a coupling constant. The curvature threshold k_{loc} induces a nontrivial background and controls the fluctuation spectrum.

3 Mass Gap Theorem

Theorem (Mass Gap in a Curvature-Threshold Scalar Topographic EFT). Consider the scalar field theory defined above. Let

$$\phi(x, t) = \phi_0(x) + \eta(x, t)$$

and quantize the quadratic fluctuation field η . Then the fluctuation spectrum possesses a strictly positive mass gap

$$m_{\text{gap}} = \sqrt{\lambda} k_{\text{loc}}.$$

Proof. Expanding around the background ϕ_0 , the quadratic Lagrangian for fluctuations is

$$L^{(2)} = \frac{1}{2}(\partial_t \eta)^2 - \frac{1}{2}(\nabla \eta)^2 - \frac{\lambda}{2}(\nabla^2 \eta)^2 - \lambda k_{\text{loc}} \eta \nabla^2 \eta - \frac{\lambda}{2} k_{\text{loc}}^2 \eta^2.$$

Integrating by parts yields the quadratic Hamiltonian

$$H^{(2)} = \frac{1}{2} \int d^3x [\pi^2 + \eta O \eta],$$

where $\pi = \partial_t \eta$ and the fluctuation operator is

$$O = -\nabla^2 + \lambda(-\nabla^2 - k_{\text{loc}})^2.$$

Acting on plane waves $e^{ik \cdot x}$, the operator has eigenvalues

$$O(k) = k^2 + \lambda(k^2 + k_{\text{loc}})^2.$$

Thus the dispersion relation is

$$\omega_k^2 = k^2 + \lambda(k^2 + k_{\text{loc}})^2.$$

Evaluating at $k = 0$ gives

$$\omega_0^2 = \lambda k_{\text{loc}}^2,$$

so the lowest excitation has mass

$$m_{\text{gap}} = \omega_0 = \sqrt{\lambda} k_{\text{loc}}.$$

Since $k_{\text{loc}} > 0$, the spectrum has a strictly positive mass gap. This completes the proof. $\square$

4 Yang–Mills–Parallel Commentary

The curvature-threshold scalar EFT mirrors the mathematical structure of a Yang–Mills mass-gap proof in four key respects:

- **Nonlinear Background:** Yang–Mills theory exhibits nontrivial gauge vacua. Here, the background ϕ_0 satisfies a curvature-activation condition $-\nabla^2\phi_0 = k_{\text{loc}} > 0$.
- **Positive Elliptic Operator:** The fluctuation operator O is self-adjoint, elliptic, and strictly positive for $k_{\text{loc}} > 0$, analogous to the gauge-covariant Laplacian.
- **Discrete Spectrum and Mass Gap:** The dispersion relation yields a strictly positive mass gap $m_{\text{gap}} = \sqrt{\lambda} k_{\text{loc}}$, structurally mirroring the conjectured Yang–Mills spectral gap.
- **Geometric Origin of Mass:** In Yang–Mills, mass arises from nonabelian field-strength geometry. Here, mass arises from internal curvature thresholds.

These parallels suggest that the scalar EFT captures the essential features of a mass-gap–generating mechanism and may serve as a tractable analogue for exploring nonperturbative spectral structure.

5 Unified Narrative for Manuscript Development

The curvature-threshold scalar topographic EFT provides a minimal, analytically transparent model in which mass arises from internal geometric structure. The model exhibits:

- a localized nonlinear background,
- a positive elliptic fluctuation operator,
- a rigorously computable spectral gap,
- and a geometric mechanism for mass generation.

These features mirror the mathematical architecture required for a Yang–Mills mass-gap proof. The framework offers a falsifiable, analytically tractable pathway for exploring mass-gap mechanisms and may serve as a geometric scaffold for future gauge-theoretic generalizations.

6 Conclusion

We have presented a scalar topographic EFT in which mass arises from a curvature-threshold mechanism. The model exhibits a rigorously demonstrable mass gap and mirrors the mathematical structure of the Yang–Mills mass-gap problem. Its simplicity, geometric clarity, and spectral positivity make it a compelling candidate for further exploration. Future work may investigate gauge-theoretic extensions, covariant formulations, and potential embeddings within nonabelian field theories.